

Data Detective

Every graded set, every week — 3 weeks, 15 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Reading a Graph

Each question describes a graph in words. Find the scale before you answer anything.

Warm-Up 6 ITEMS

Each ★ = 5 books. Monday has 3 ★ — books

Each ★ = 5 books. Friday has 6 ★ — books

A bar reaches 40 on a scale marked every 10 — its value

Each ■ = 2 goals. A bar of 7 ■ — goals

Bars: Red 8, Blue 5 — total

Bars: Red 8, Blue 5 — difference

Core GRADED • SHOW YOUR WORKING

1. Scale every 5. A bar halfway between 20 and 30 (Halfway is one mark up)

2. Three full groups of five tallies plus 2 — total (15 and 2 more)

3. Graph shows Mon 12, Tue 9, Wed 15 — three-day total

4. Each ■ = 2 goals. 24 goals is how many ■ (24 ÷ 2)

5. Each ★ = 4 pets. 6 ★ and a half ★ — pets (Half a star is 2)

★ Challenge NOT GRADED

C1. Mon 12, Tue 9, Wed 15, Thu 0 — mean of the four days (36 ÷ 4 — the zero still counts as a day)

C2. Each ★ = 5. Four rows total 90 books — stars altogether (90 ÷ 5)

Mean, Median, Mode, Range

Every set here is chosen so the answer comes out whole. Put the numbers in order before you look for a middle.

Warm-Up 6 ITEMS

Mean of 4, 6, 8

Mean of 10 and 20

Range of 3, 9, 12

Mode of 5, 7, 7, 9

Median of 2, 5, 9

Total of 6, 7, 8, 9

Core GRADED • SHOW YOUR WORKING

1. Mean of 4, 7, 7, 9, 13 (They total 40)

.....

2. Median of 4, 7, 7, 9, 13

.....

3. Range of 4, 7, 7, 9, 13 (13 - 4)

.....

4. Median of 2, 4, 6, 8 (Halfway between the two middles)

.....

5. Mean of 12, 14, 16, 18

.....

Challenge NOT GRADED

C1. Five numbers have mean 10 — their total (Mean $\times$ how many)

.....

C2. Four scores total 32. Add a fifth score of 8 — new mean (40 $\div$ 5)

.....

When the Average Lies

One extreme value. Watch carefully which measures move and which do not.

 Warm-Up 6 ITEMS

Mean of 5, 5, 5, 5 _____	Median of 5, 5, 5, 5 _____	Mean of 5, 5, 5, 25 _____
Median of 5, 5, 5, 25 _____	Range of 5, 5, 5, 25 _____	Mode of 5, 5, 5, 25 _____

 Core GRADED • SHOW YOUR WORKING

1. Four people earn 10 each, a fifth earns 60 — mean ($100 \div 5$)	_____
2. Those same five people — median	_____
3. Those same five people — range	_____
4. Remove the 60. Mean of the four left	_____
5. Type the value of the measure that best describes a typical one of those five people (The median, 10 — the mean of 20 describes nobody)	_____

 Challenge NOT GRADED

C1. Six houses cost 200 each and one costs 900 — mean ($2100 \div 7$)	_____
C2. Those same seven houses — median	_____

Chance as a Number

Chance is a count out of a total. Type the count that is asked for.

✦ Warm-Up 6 ITEMS

Faces on one die

Ways to roll a 4 on one die

Even numbers on one die

Ways to roll more than 4 on one die

Bag of 3 red and 5 blue — total counters

That bag — how many draws out of 8 should be red

◆ Core GRADED • SHOW YOUR WORKING

1. Chance of an even roll is _ out of 6 — type the top number

.....

2. 60 rolls of a die — expected number of 3s (60 ÷ 6)

.....

3. Bag of 3 red, 5 blue. 40 draws, replacing each time — expected reds (3 in every 8, and 40 is five eights)

.....

4. A coin flipped 50 times — expected heads

.....

5. Spinner in 8 equal parts, 2 gold. 40 spins — expected golds (2 in every 8)

.....

★ Challenge NOT GRADED

C1. Two dice — how many ways to total 7 (1+6, 2+5, 3+4 and each one reversed)

.....

C2. Two dice — how many outcomes are possible altogether (6 for the first die, 6 for the second)

.....

Predict the Roll & Ask 20 People

Enrichment day. Predict first, in writing, every single time.

✦ Warm-Up 2 ITEMS

People in the survey

Trials in Predict the Roll

◆ Core GRADED • SHOW YOUR WORKING

1. 20 people, 15 say yes — percent saying yes (15 out of 20 is 75 out of 100)

.....

2. 20 people, 4 pick blue. Out of 100 people, expect (Multiply both by 5)

.....

★ Challenge NOT GRADED

C1. 50 rolls of a die — expected number of 6s (50 ÷ 6 is 8 with 2 over)

C2. 20 people own a mean of 6 pets each — total pets

C3. You predicted 8 sixes and rolled 14. Difference (Real data misses the prediction. That is not an error.)

Week 1 Answers

Mission 07 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 • Reading a Graph

OUT OF 11

Warm-Up • 15 | 30 | 40 | 14 | 13 | 3

Core • 25 | 17 | 36 | 12 | 26

Challenge • 9 | 18

1.2 • Mean, Median, Mode, Range

OUT OF 11

Warm-Up • 6 | 15 | 9 | 7 | 5 | 30

Core • 8 | 7 | 9 | 5 | 15

Challenge • 50 | 8

1.3 • When the Average Lies

OUT OF 11

Warm-Up • 5 | 5 | 10 | 5 | 20 | 5

Core • 20 | 10 | 50 | 10 | 10

Challenge • 300 | 200

1.4 • Chance as a Number

OUT OF 11

Warm-Up • 6 | 1 | 3 | 2 | 8 | 3

Core • 3 | 10 | 15 | 25 | 10

Challenge • 6 | 36

Fri • Predict the Roll & Ask 20 People

OUT OF 4

Warm-Up • 20 | 50

Core • 75 | 20

Challenge • 8 | 120 | 6

Write the Question

A fair question doesn't push the answer. Rewrite the loaded ones.

Warm-Up 6 ITEMS

People you will ask

Answer choices in a yes/no question

20 people, 10 say yes — percent
saying yes

20 people, 5 say yes — percent

20 people, 20 say yes — percent

20 people, 0 say yes — percent

Core GRADED • SHOW YOUR WORKING

1. 20 people, 15 say yes — percent

.....

2. 20 people, 12 say yes — percent

.....

3. 20 people, 3 say yes — percent

.....

4. If 60% of 20 said yes, how many people

.....

5. 20 people across 4 choices, evenly — each choice

.....

Challenge NOT GRADED

C1. 20 people, 7 say yes — percent

.....

C2. Out of 100 people, expect how many yes if 7 of 20 said yes

.....

Ask 20 People

Tally as they answer. No tidying up afterwards.

Warm-Up 6 ITEMS

Four tally marks plus a cross-stroke — the count

Two groups of five

Three groups of five plus 2

Four groups of five

One group of five plus 3

20 minus 13

Core GRADED • SHOW YOUR WORKING

1. Counts 8, 6, 4, 2 — the total

.....

2. Those counts — the largest

.....

3. Those counts — the range

.....

4. Counts 9, 5, 6 — the total

.....

5. If three choices total 20 and two are 9 and 5 — the third

.....

Challenge NOT GRADED

C1. 20 answers over 4 choices, one has 11 — the other three total

.....

C2. Counts 8, 6, 4, 2 — the mean

.....

Graph What You Got

Bar graph of your own data, plus all four measures.

Warm-Up 6 ITEMS

Counts 8, 6, 4, 2 — the total

Their mean

Their range

The mode of 5, 7, 7, 9

The median of 2, 5, 9

A scale marked every 2 — the value 3 marks up

Core GRADED • SHOW YOUR WORKING

1. Counts 12, 8, 6, 4 — the mean

2. Those counts — the median

3. Those counts — the range

4. Water 8, juice 6, milk 4, tea 2 — the mode

5. Out of 100 people, expect how many water

Challenge NOT GRADED

C1. Counts 9, 5, 3, 2, 1 — the mean

C2. Those counts — the median

Chance as a Number

Chance is a count out of a total. Nothing more mysterious.

✦ Warm-Up 6 ITEMS

Faces on one die

Ways to roll a 4

Even numbers on a die

Ways to roll more than 4

A bag of 3 red and 5 blue — total

Sides on a coin

◆ Core GRADED • SHOW YOUR WORKING

1. 60 rolls of a die — expected 3s

.....

2. A coin flipped 50 times — expected heads

.....

3. 3 red in 8, drawn 40 times — expected reds

.....

4. A spinner of 8 parts, 2 gold, 40 spins — expected golds

.....

5. A bag of 4 red and 6 blue — reds expected in 20 draws

.....

★ Challenge NOT GRADED

C1. Two dice — ways to total 7

.....

C2. Two dice — total possible outcomes

.....

Mid-Unit Quiz

Eight items across Weeks 1–2. 85% to keep flying.

✦ Warm-Up 2 ITEMS

Each ★ = 5 books. 7 ★ — books

Mean of 6, 8, 10, 12

◆ Core GRADED • SHOW YOUR WORKING

1. Median of 11, 4, 7, 20, 9

.....

2. Mode and range of 3, 8, 8, 8, 15 — type the range

.....

3. Six numbers with mean 9 — their total

.....

4. A bag of 4 red and 6 blue — reds in 10 draws

.....

5. 60 rolls — expected 5s

.....

★ Challenge NOT GRADED

C1. 2, 2, 2, 2, 22 — the mean

.....

Week 2 Answers

Mission 07 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 • Write the Question

OUT OF 11

Warm-Up • 20 | 2 | 50 | 25 | 100 | 0

Core • 75 | 60 | 15 | 12 | 5

Challenge • 35 | 35

2.2 • Ask 20 People

OUT OF 11

Warm-Up • 5 | 10 | 17 | 20 | 8 | 7

Core • 20 | 8 | 6 | 20 | 6

Challenge • 9 | 5

2.3 • Graph What You Got

OUT OF 11

Warm-Up • 20 | 5 | 6 | 7 | 5 | 6

Core • 7.5 | 7 | 8 | water | 40

Challenge • 4 | 3

2.4 • Chance as a Number

OUT OF 11

Warm-Up • 6 | 1 | 3 | 2 | 8 | 2

Core • 10 | 25 | 15 | 10 | 8

Challenge • 6 | 36

Fri • Mid-Unit Quiz

OUT OF 7

Warm-Up • 35 | 9

Core • 9 | 12 | 54 | 4 | 10

Challenge • 6

Publish the Findings

One headline, one graph, one honest limitation.

Warm-Up 6 ITEMS

People surveyed _____	15 of 20 — the percent _____	5 of 20 — the percent _____
Out of 100, expect how many if 15 of 20 said yes _____	8 of 20 — the percent _____	20 minus 8 _____

Core GRADED • SHOW YOUR WORKING

1. 12 of 20 — out of 100
.....

2. A headline says most people. The smallest count of 20 that earns it
.....

3. 9 of 20 — the percent
.....

4. Does 45% earn most — yes or no
.....

5. 17 of 20 — the percent
.....

Challenge NOT GRADED

C1. A survey of 20 predicting a town of 5,000 — people per person surveyed
.....

C2. If 15 of 20 said yes, the predicted yes count in 5,000
.....

The Misleading Graph

Start the axis at 90 and watch a tiny gap look enormous.

Warm-Up 6 ITEMS

Bars at 95 and 98 — the true difference

Bars at 50 and 60 — the difference

An axis starting at 0 shows a bar of 95 as how tall, in units

An axis starting at 90 shows that bar as

Bars at 20 and 40 — the difference

40 is how many times 20

Core GRADED • SHOW YOUR WORKING

1. Bars at 95 and 98 on an axis from 90 — the drawn heights differ by

.....

2. Drawn from 90, 98 appears how many times taller than 95

.....

3. The true ratio of 98 to 95, to one decimal

.....

4. An axis starting at 0 is honest — yes or no

.....

5. Bars at 102 and 104 from an axis at 100 — apparent times taller

.....

Challenge NOT GRADED

C1. Sales of 1,020 and 1,040 drawn from 1,000 — apparent times taller

.....

C2. Their true percent difference, to the nearest whole

.....

Fifty Trials

Predict, roll, tally, and explain the gap in writing.

Warm-Up 6 ITEMS

Trials

Faces on a die

$50 \div 6$ — the whole part

$50 \div 6$ — the remainder

A coin, 50 flips — expected heads

Six faces, expected each in 60 rolls

Core GRADED • SHOW YOUR WORKING

1. Predicted 8, rolled 14 — the difference

2. Predicted 8, rolled 3 — the difference

3. 600 rolls — expected 6s

4. Is a spread of a few either way normal in 50 rolls — yes or no

5. Six tallies totalling 50, five are 8 — the sixth

Challenge NOT GRADED

C1. Two dice, 36 rolls — expected 7s

C2. Two dice — ways to total 2

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

✦ Warm-Up 6 ITEMS

Mean of 4, 6, 8

Median of 2, 5, 9

Mode of 5, 7, 7, 9

Range of 3, 9, 12

Each ★ = 5. 6 ★ — books

Even numbers on a die

◆ Core GRADED • SHOW YOUR WORKING

1. Mean of 4, 7, 7, 9, 13

.....

2. Median of that set

.....

3. Five numbers with mean 10 — their total

.....

4. 60 rolls — expected 3s

.....

5. 15 of 20 — the percent

.....

★ Challenge NOT GRADED

C1. Six houses at 200 and one at 900 — the mean

.....

C2. Those same houses — the median

.....

Mission 07 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 2 ITEMS

A bar chart marked every 4, a bar at the third mark — its value

Each ■ = 2 goals. 9 ■ — goals

Core GRADED • SHOW YOUR WORKING

1. Mean of 14, 16, 18, 20

.....

2. Median of 5, 12, 3, 8, 21, 9

.....

3. Range of 7, 7, 19, 4, 11

.....

4. Five scores total 45, a sixth of 9 added — new mean

.....

5. A spinner of 8 parts, 2 gold, 40 spins — expected golds

.....

6. 20 people, 8 pick water — out of 100

.....

7. Water 8, juice 6, milk 4, tea 2 — the mean

.....

8. Those four — the mode

.....

Challenge NOT GRADED

C1. Two dice — ways to total 7

C2. 2, 2, 2, 2, 22 — mean and median; type the median

Week 3 Answers

Mission 07 • all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 • Publish the Findings

OUT OF 11

Warm-Up • 20 | 75 | 25 | 75 | 40 | 12

Core • 60 | 11 | 45 | no | 85

Challenge • 250 | 3750

3.2 • The Misleading Graph

OUT OF 11

Warm-Up • 3 | 10 | 95 | 5 | 20 | 2

Core • 3 | 1.6 | 1 | yes | 2

Challenge • 2 | 2

3.3 • Fifty Trials

OUT OF 11

Warm-Up • 50 | 6 | 8 | 2 | 25 | 10

Core • 6 | 5 | 100 | yes | 10

Challenge • 6 | 1

Thu • Error Journal Sweep

OUT OF 11

Warm-Up • 6 | 5 | 7 | 9 | 30 | 3

Core • 8 | 7 | 50 | 10 | 75

Challenge • 300 | 200

Fri • Mission 07 Test

OUT OF 10

Warm-Up • 12 | 18

Core • 17 | 8.5 | 15 | 9 | 10 | 40 | 5 | water

Challenge • 6 | 2